

Pari Jain

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Profile Summary

Second-year B.Tech IT student interested in Large Language Models, knowledge-grounded dialogue systems, retrieval-augmented generation, and AI for healthcare. Currently exploring how LLMs can reason over evolving user data using retrieval, embeddings, and structured knowledge. Long-term vision includes integrating AI with the fashion industry through intelligent personalization, recommendation, and context-aware design systems.

Education

Swami Keshvanand Institute of Technology, Jaipur

2024 – 2028

B.Tech in Information Technology

Research-Oriented Projects

Smart Community Health Monitoring & Early Warning System for Water-Borne Diseases

- Developed a digital twin-based health monitoring system to predict water-borne disease outbreaks in rural communities.
- Applied computer vision techniques to analyze images for detecting water contamination and sanitation risks.
- Integrated environmental and health datasets to generate early warning alerts for proactive disease prevention.
- Focused on community-level risk assessment and scalable monitoring for public health applications.

Knowledge-Grounded Healthcare Dialogue System

- Designed a prototype healthcare dialogue system that retrieves relevant medical context before generating responses, reducing dependence on purely generative outputs.
- Used sentence-transformer embeddings and FAISS-based similarity search to retrieve symptom-related information for context-aware response generation.
- Explored knowledge-grounding concepts to improve factual consistency and reduce hallucination in LLM-style healthcare interactions.

Temporal Digital Twin for Personalized Health Tracking

- Built a health-focused digital twin concept that maintains evolving user profiles using symptoms, lifestyle patterns, prediction history, and temporal health records.
- Integrated embedding-based memory representation to retrieve similar past health states and support longitudinal health analysis.
- Focused on adaptive reasoning over changing user data instead of static one-time disease prediction.

Current Research Focus

- Exploring knowledge-grounded and retrieval-based dialogue systems for healthcare, focusing on how external context can improve LLM response reliability.
- Studying retrieval-augmented generation architectures for reducing hallucinations and improving factual consistency in AI-generated medical insights.
- Interested in LLM reasoning over evolving user profiles, especially combining temporal data, structured health records, and unstructured symptom descriptions.
- Exploring multilingual NLP and Indian-language AI systems for making healthcare dialogue systems more accessible in diverse linguistic settings.

Technical Skills

- **Languages:** Python, C++, JavaScript
- **Machine Learning:** Supervised Learning, Feature Engineering, Model Evaluation
- **NLP & LLMs:** Text Embeddings, Sentence Transformers, Tokenization, Prompting Basics
- **Retrieval Systems:** FAISS, Embedding-Based Similarity Search, RAG Concepts
- **Libraries/Tools:** NumPy, Pandas, Scikit-learn, Matplotlib, Flask
- **Databases/Web:** MySQL, HTML, CSS, JavaScript

Research Interests

- Knowledge-Grounded Dialogue Systems
- Retrieval-Augmented Generation
- LLM Reasoning and Interpretability
- Healthcare NLP and Medical Question Answering
- Multilingual NLP for Indian Languages
- Digital Twin Systems for Personalized Healthcare

Ongoing Exploration

- Currently experimenting with embedding-based similarity search using FAISS and sentence-transformers to simulate retrieval behavior in evolving datasets.
- Exploring how knowledge-grounded dialogue systems can combine retrieved context with user-specific temporal health data for personalized AI assistance.
- Studying how LLM-based systems can be made more reliable through retrieval, grounding, and interpretable reasoning.

Certifications

- Machine Learning Specialization – Coursera / Stanford Online / DeepLearning.AI
- Introduction to Generative AI – Google Cloud
- Introduction to Large Language Models – Google Cloud
- Introduction to Responsible AI – Google Cloud
- Introduction to Artificial Intelligence – IBM
- HTML5, CSS3, JavaScript – Infosys Springboard